



# EDUPATH

-a personalized study guidance system



**Project Guide:**  
Leya G

| Roll No | KTU Registration No | Name                   |
|---------|---------------------|------------------------|
| 17      | CHN23CS055          | ATHIRA MP              |
| 34      | CHN23CS105          | JOSEPH<br>GREGORY BIJU |
| 62      | CHN23CS182          | SUHANA K               |
| 67      | LCHN23CS199         | AKASH N                |



# INTRODUCTION

Engineering students often struggle to manage vast syllabi and exam pressure due to limited time and the lack of personalized study guidance. This system aims to assist students by generating personalized study plans tailored to their learning needs.

- Analyzes student performance through assessments.
- Determines the student's knowledge level (Basic / Intermediate / Advanced).
- Considers available study time and exam schedules.
- Recommends appropriate learning resources and study plans.



# PROBLEM STATEMENT

Engineering students face difficulties in managing large syllabi, identifying appropriate learning resources, and planning effective study schedules due to the lack of personalized guidance and self-assessment mechanisms.

# LITERATURE REVIEW

| SI No | Name of paper                                                                      | Year | Methodology                                                                                                                                                                         | Result                                                             | Limitation                                                                      |
|-------|------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1.    | Personalized Dynamic Learning Plan Generator for Smart Learning Environments       | 2019 | Uses Dynamic Learning Plan Selector (DLPS) and Dynamic Learning Plan Generator (DLPG) algorithms to create personalized learning schedules based on learner progress and deadlines. | Adaptive learning plans, course completion and learner motivation. | Requires accurate learner data and integration with real-time learning systems. |
| 2.    | Assessment-Centered E-Learning system for improving student learning effectiveness | 2013 | Web-based diagnostic assessment system using personalized dynamic assessments and adaptive learning materials.                                                                      | Improved system learning achievement and reduced misconceptions.   | Requires well-designed assessment items and accurate evaluation mechanisms.     |

# LITERATURE REVIEW

| SI No | Name of paper                                            | Year | Methodology                                                                                                                        | Result                                                                                           | Limitation                                                                       |
|-------|----------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 3.    | A Study of Internet Instant Messaging and Chat Protocols | 2006 | Public chat platform based on client-server architecture allowing users to communicate and exchange messages in real time.         | Identifies common messaging features such as chat rooms, authentication and messaging protocols. | Security vulnerabilities and lack of protocol standardization limit scalability. |
| 4.    | Adaptive E-Learning                                      | 2003 | Uses Aptitude-Treatment Interaction (ATI) theory to personalize learning content based on learner characteristics and performance. | Personalized content improves learning effectiveness and engagement.                             | Designing accurate learner model is complex and costly.                          |

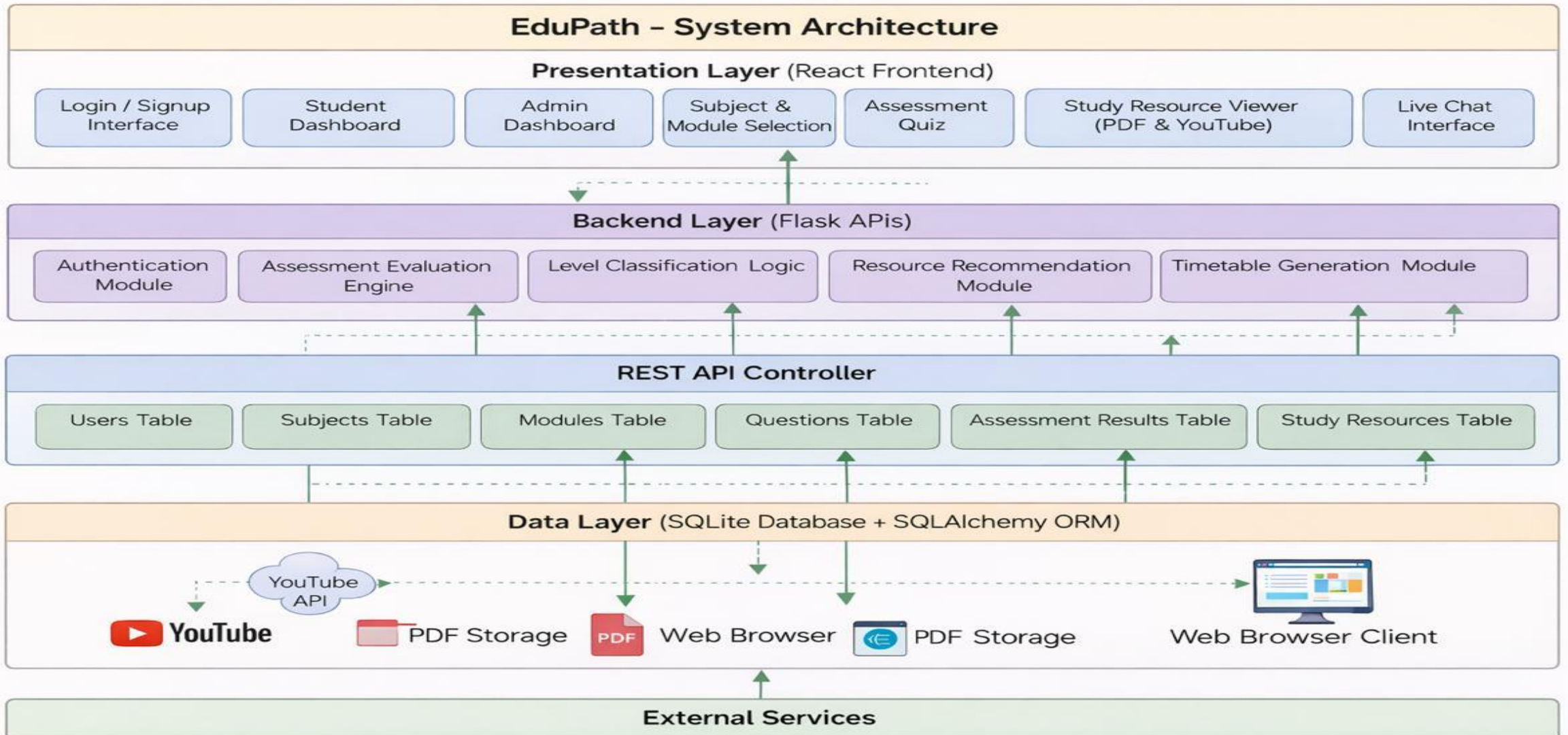


# PROPOSED SOLUTION

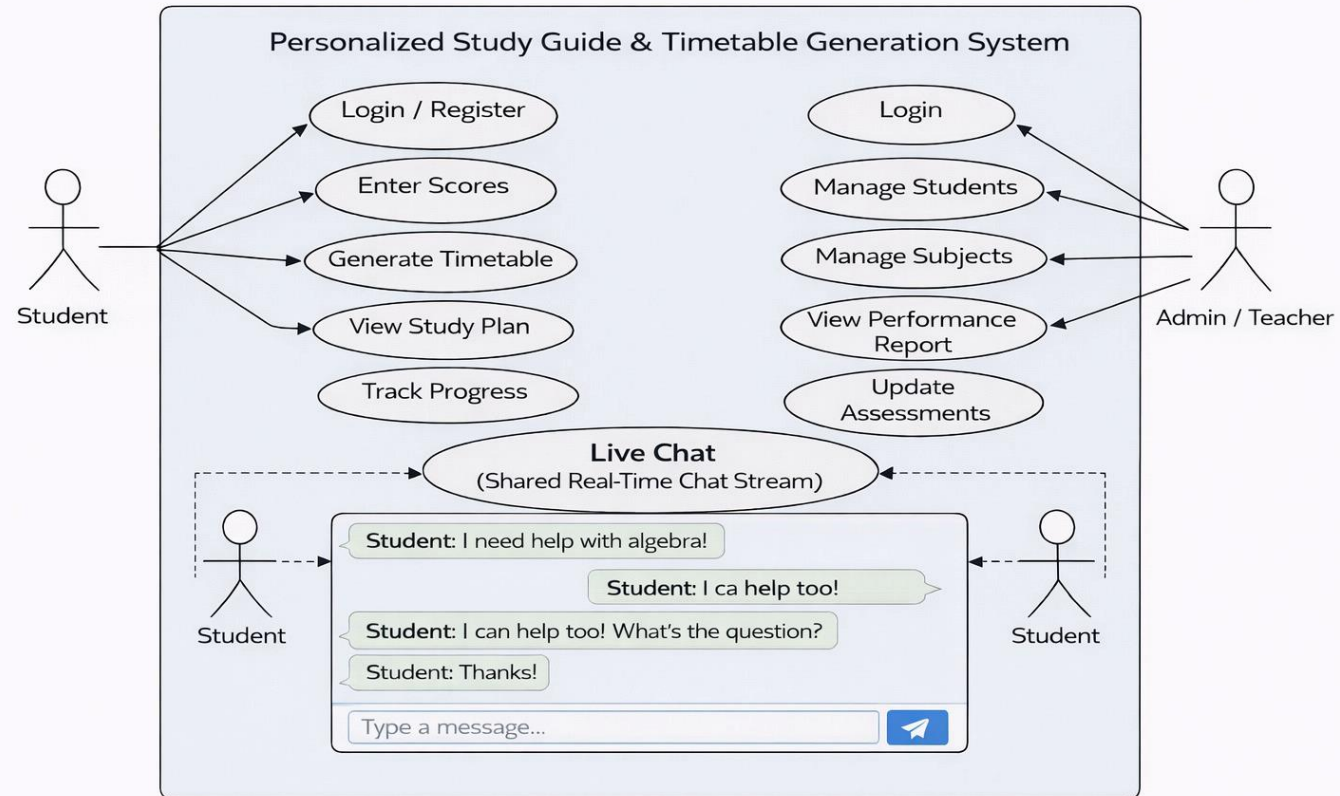
- Collects student details, subjects, and performance data
- Identifies individual strengths and weaknesses
- Allocates study time according to availability
- Develop a Personalized Study Planning System



# SYSTEM ARCHITECTURE



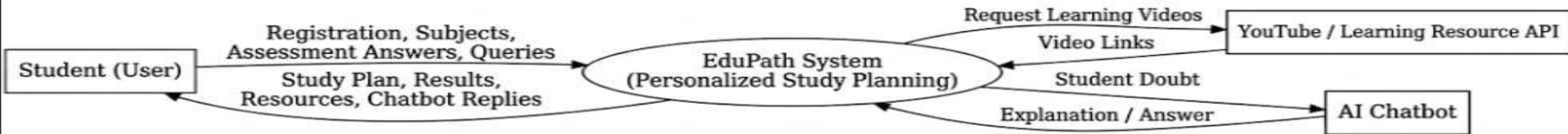
# USE CASE DIAGRAM



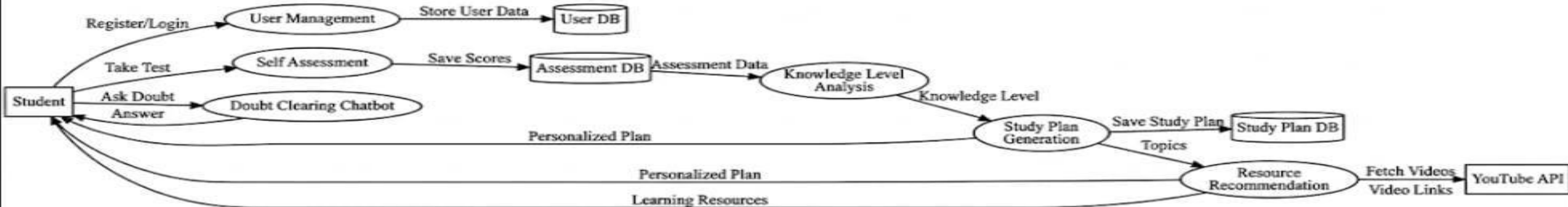


# DATA FLOW DIAGRAM

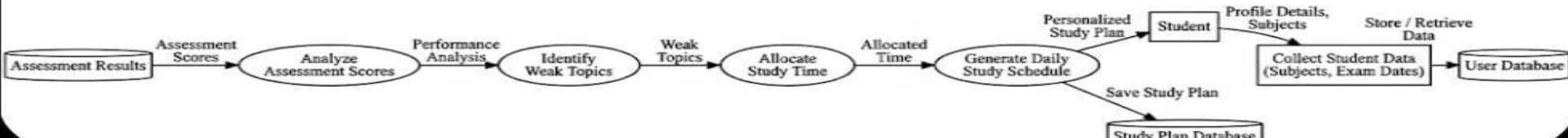
## Level 0 DFD: EduPath System Overview



## Level 1 DFD: System Sub-processes



## Level 2 DFD: Study Plan Generation Process





# METHODOLOGY

## 1. Requirement Analysis

Identified system requirements for students and admin.

Defined features such as quiz system, timetable generation, learning resources, and chat box interaction.

## 2. System Design

Designed client–server architecture.

Created database structure for users, subjects, modules, quizzes, and chat messages.

Designed user interface for login, quizzes, timetable, and chat.

## 3. System Development

Frontend: Developed using React, HTML, and CSS.

Backend: Implemented using Flask (Python).



# METHODOLOGY

Implemented features such as authentication, quiz evaluation, and resource redirection.

## **4. Database Implementation**

Stored user profiles, subjects, modules, quiz questions, results, and chat messages.

Ensured efficient data storage and retrieval.

## **5. System Integration**

Integrated frontend, backend, and database.

Enabled communication through API requests and responses.

## **6. Timetable Generation Logic**

Generated personalized study timetable.

Allocated study time based on student knowledge level (Basic, Intermediate, Advanced).



# METHODOLOGY

## **7. Chat box Implementation**

Implemented real-time chat system.

Enabled students to communicate and discuss study topics.

## **8. Testing and Deployment**

Tested login, quiz evaluation, timetable generation, and chat functionality.

Deployed the system on local server for user access.



# FUNCTIONAL REQUIREMENTS

## **User and Admin Authentication**

The system must provide two separate login interfaces: one for Admin and one for Users (Students).

Users must be able to register and log in securely using their credentials.

## **Subject and Module Management (Admin)**

The admin must be able to add subjects to the system.

The admin must be able to create and delete modules for each subject.

## **Quiz and Assessment System**

The admin must be able to add quiz questions and answer options for each module.

The system must conduct quizzes, evaluate student responses automatically and store the results.





# FUNCTIONAL REQUIREMENTS

## **Knowledge Level Identification**

The system must analyze the student's quiz performance.

Based on the results, the system must classify the students into:  
“Basic”, “Intermediate”, “Advanced”

## **Learning Resource Recommendation**

According to the identified knowledge level, the system must redirect students to appropriate learning resources such as:

- ✓ PDF study materials
- ✓ YouTube video tutorials

## **Real-Time Chat box Communication**

Users must be able to:

- ✓ Send messages
- ✓ View messages from other users instantly





# FUNCTIONAL REQUIREMENTS

## **Personalized Timetable Generation**

The system must generate a personalized study timetable for each student.

The timetable must allocate different study durations based on the student's knowledge level:

Basic level: More time allocated for studying the subject

Intermediate level: Moderate study time allocated

Advanced level: Less study time allocated



# NON FUNCTIONAL REQUIREMENTS

## **Performance**

Fast response for login, assessments, and resource loading.

## **Security**

Secure user authentication and protected data storage.

## **Usability**

Simple and user-friendly interface.

## **Scalability**

Supports multiple users and chat interactions.



# NON FUNCTIONAL REQUIREMENTS

## **Reliability**

Accurate assessment evaluation and level classification.

## **Availability**

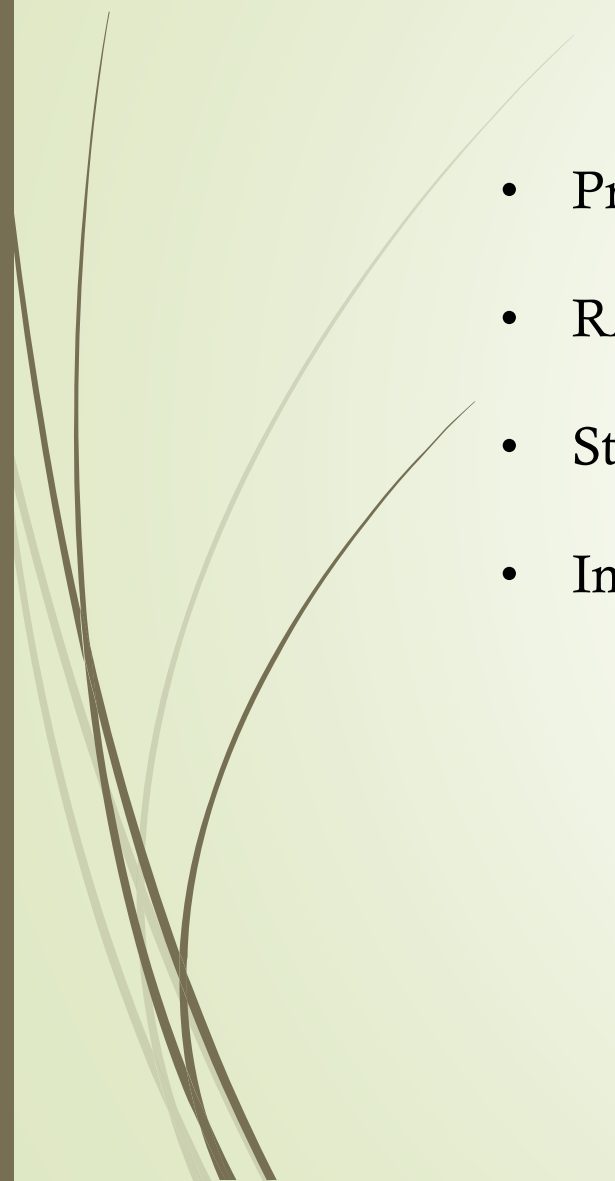
System accessible anytime through a web browser.

## **Maintainability**

Admin can easily update subjects, modules, and questions.



# HARDWARE REQUIREMENTS

- Processor: Intel i3 or above
  - RAM: Minimum 4 GB
  - Storage: Minimum 10 GB free disk space
  - Internet connection required
- 



# SOFTWARE REQUIREMENTS

- Frontend: React.js
- Backend: Python (Flask)
- Database: SQLite
- Operating System: Windows / Linux





# RESULT

- Developed EduPath – a personalized study guidance system for students.
- Students can take assessment quizzes to evaluate their knowledge level.
- The system classifies learners into Basic, Intermediate, or Advanced levels.
- Provides level-based learning resources such as PDF notes and YouTube videos.
- Enables real-time communication through a chat feature.
- Admin can manage subjects, modules, questions, and study materials.
- Helps students organize their studies and prepare effectively for exams.

## Join EduPath

Full Name

John Doe

Email Address

student@example.com

Password

.....

Account Type

Student



Student

Administrator

Already have an account? [Sign in](#)

# Welcome to EduPath

Email Address

joseph@gmail.com

Password

.....|



Sign In

Don't have an account? [Sign up](#)

## Admin Portal

 Manage Subjects

 Manage Modules

 Add Questions

 Input Notes (PDF/YT)

 Student Performance

 Log Out

Ad  
Log

A

localhost:5173 says

Subject created successfully!

OK

### Manage Subjects & Notes Link

Subject Name

AAD

Exam Date

07-04-2026



Add Subject

### Existing Subjects

| ID | Name | Exam Date  |
|----|------|------------|
| 1  | AAD  | 2026-04-07 |

# Admin Portal

 Manage Subjects

 Manage Modules

 Add Questions

 Input Notes (PDF/YT)

 Student Performance

 Log Out

## Administrator controls

Logged in as akashnandan227@gmail.com

A

### Create Module / Chapter

#### Select Subject

AAD

#### Module Title

e.g. Chapter 1: Sorting Algorithms

Add Module

### Modules for: AAD

| ID | Module Name | Action                                                                                       |
|----|-------------|----------------------------------------------------------------------------------------------|
| 1  | MODULE 1    |  Delete |

## Admin Portal

 Manage Subjects

 Manage Modules

 Add Questions

 Input Notes (PDF/YT)

 Student Performance

 Log Out

### Add Quiz Questions

Subject

AAD

Module

MODULE 1

Question Text

What is the time complexity of QuickSort?

Difficulty Level

Basic

Option A

Option B

Option C

Option D

Correct Option



## Admin Portal

 Manage Subjects

 Manage Modules

 Add Questions

 Input Notes (PDF/YT)

 Student Performance

 Log Out

### Upload Reference Notes & Media

Subject

AAD

Module

MODULE 1

Target Level

Basic Level Students

PDF Notes Upload (Optional)

No file chosen

YouTube Video Link (Optional)

<https://youtube.com/watch?v=...>

Save Notes Content

## Student Portal

 Profile & Dashboard

 Time Table

 Knowledge  
Assessment

 Live Chat Box

 Log Out



**Name:** joseph

**Email:** joseph@gmail.com

**College:** College Of Engineering Chengannur

**Role:** Student Status

### BROWSE SUBJECTS

AAD

 2026-04-07

Start Full Quiz →

CG

 2026-04-17

Start Full Quiz →

PYTHON

 2026-04-22

Start Full Quiz →

ML

 2026-04-24

Start Full Quiz →

AAD

 2026-04-07

Start Full Quiz →

## Student Portal

 Profile & Dashboard

 Time Table

 Knowledge  
Assessment

 Live Chat Box

 Log Out

### Module Quiz — MODULE 1

1 / 5

MODULE 1

MODULE 2

MODULE 3

MODULE 4

MODULE 5

Knowledge Assessment Quiz

20 Questions

#### 1. What is Computer Graphics?

☐ Drawing pictures using a computer

☐ Managing databases

☐ Network communication

☐ Writing only text programs

#### 2. The smallest unit of an image is called:

☐ Pixel

☐ Byte

☐ Frame

☐ Bit

## Student Portal

 Profile & Dashboard

 Time Table

 Knowledge  
Assessment

 Live Chat Box

 Log Out

## All Modules Complete!

Here's a summary of your performance across all modules.

### MODULE 1

Basic: 3 | Inter: 1 | Adv: 3

 Basic

### MODULE 2

Basic: 5 | Inter: 2 | Adv: 6

 Intermediate

### MODULE 3

Basic: 4 | Inter: 5 | Adv: 4

 Advanced

### MODULE 4

Basic: 1 | Inter: 3 | Adv: 2

 Basic

### MODULE 5

Basic: 2 | Inter: 2 | Adv: 0

 Basic

[Back to Home](#)

[View Timetable →](#)

# Student Portal

 Profile & Dashboard

 Time Table

 Knowledge  
Assessment

 Live Chat Box

 Log Out

## Smart Study Timetable

Your dynamic study plan based on assessment results. Each subject shows per-module notes and YouTube links.

**AAD**

 Exam: 2026-04-07

DAYS LEFT  
**29**

Basic

DAILY  
**3h**

5 modules with  
resources

▲ Hide Notes

MODULE 1 **Advanced**

 PDF Notes

 YouTube

MODULE 2 **Basic**

 PDF Notes

 YouTube

MODULE 3 **Basic**

 PDF Notes

 YouTube

MODULE 4 **Basic**

 PDF Notes

 YouTube

MODULE 5 **Intermediate**

 PDF Notes

 YouTube

**CG**

 Exam: 2026-04-17

DAYS LEFT  
**39**

Basic

DAILY  
**3h**

0 modules with  
resources

▼ Show Notes

## Student Portal

 Profile & Dashboard

 Time Table

 Knowledge  
Assessment

 Live Chat Box

 Log Out

## Live Student Chat Room

ATHIRA M P • 04:25 am  
hloo

johny • 12:19 pm  
helloo

hii

hiiii

What's up

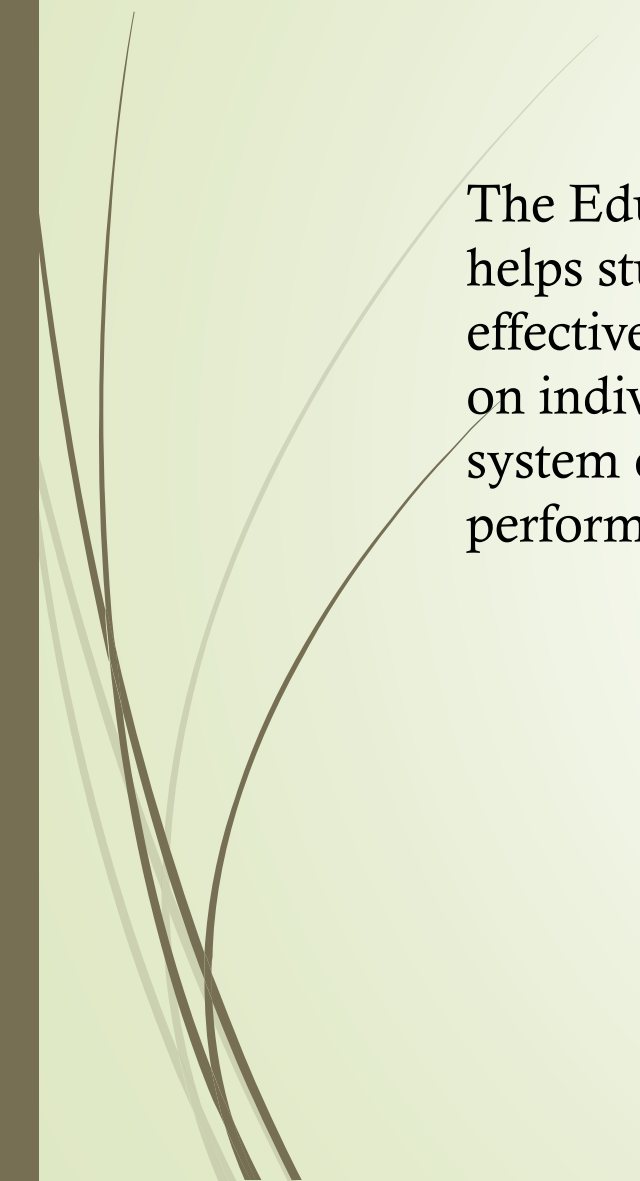
Type your message to the class...

Send



# CONCLUSION

The EduPath- Personalized Study Planning System helps students manage their academic workload effectively by generating customized study plans based on individual performance and time availability. The system enhances learning efficiency and academic performance.







# FUTURE SCOPE

- Implement progress tracking to monitor student learning performance and completion of study plans.
- Integrate an AI-based chatbot to provide instant doubt clarification and learning support.
- Develop a dynamic study plan generator that automatically updates schedules based on student progress and exam dates.
- Add advanced analytics to identify weak topics and recommend targeted learning resources.
- Extend the system to support mobile applications for easier access.
- Improve the platform by integrating more educational resources and interactive learning tools.



# REFERENCES

[https://www.sciencedirect.com/org/science/article/abs/pii/S1947851822000126?utm\\_source=google.com](https://www.sciencedirect.com/org/science/article/abs/pii/S1947851822000126?utm_source=google.com)

[https://link.springer.com/article/10.1186/s41239-021-00289-4?utm\\_source=google.com](https://link.springer.com/article/10.1186/s41239-021-00289-4?utm_source=google.com)